

Consider an x_0 and y_0 , feasible in the primal and dual respectively. That is, $Ax_0 \leq b$ and $A^T y_0 = c$; $y_0 \geq 0$. Then $c^T x_0 = b^T y_0$ if and only if $(y_0)_i > 0 \Rightarrow A_i x_0 = b_i$.

Proof: First assume that the complementary slackness condition holds. We need to prove that the costs of the points are equal, i.e., suppose $(y_0)_i > 0 \Rightarrow A_i x_0 = b_i$. To show $c^T x_0 = b^T y_0$

We know: $c_1 = a_{11}(y_0)_1 + a_{21}(y_0)_2 + \dots + a_{m1}(y_0)_m$.

$$c_2 = a_{12}(y_0)_1 + a_{22}(y_0)_2 + \dots + a_{m2}(y_0)_m$$

.

$$c_n = a_{1n}(y_0)_1 + a_{2n}(y_0)_2 + \dots + a_{mn}(y_0)_m$$

$$c^T x_0 = c_1(x_0)_1 + c_2(x_0)_2 + \dots + c_n(x_0)_n$$

$$c^T x_0 = (y_0)_1 A_1 x_0 + (y_0)_2 A_2 x_0 + \dots + (y_0)_m A_m x_0$$

By hypothesis, whenever $(y_0)_i > 0$, $A_i x_0 = b_i$

$$\therefore c^T x_0 = b^T y_0$$

Now assume that the costs are equal.

Suppose $c^T x_0 = b^T y_0$. To show $(y_0)_i > 0 \Rightarrow A_i x_0 = b_i$

Once again use the fact that:

$$c^T x_0 = (y_0)_1 A_1 x_0 + (y_0)_2 A_2 x_0 + \dots + (y_0)_m A_m x_0$$

If for some i with $(y_0)_i > 0$, we have $A_i x_0 < b_i$. Then $c^T x_0 < b^T y_0$ (Contradicting our hypothesis).

$\Rightarrow$ whenever $(y_0)_i > 0$, we have $A_i x_0 = b_i$.

Mixed Constraints:

Now consider the general problem, in which we have equality and inequality constraints:

$$\max_{u_1 \in R} f(u_1, \dots, u_n)$$

Subject to

$$g_1(u_1, u_2, \dots, u_n) \leq b_1, \dots, g_k(u_1, \dots, u_n) \leq b_k,$$

$$h_1(u_1, u_2, \dots, u_n) = c_1, \dots, h_m(u_1, \dots, u_n) = c_m.$$

Summarizing the conditions for equality and inequality constraints that we found above we can formulate the following theorem:

Theorem: Suppose that u^* is a local maximizer of f on the constraint set given by the k inequality and m equalities above. Without loss of generality, assume that the first k_0 inequality constraints are binding and that the other $k - k_0$ inequality constraints are not binding. Further suppose that the Jacobian of the equality constraints and the binding inequality constraints at u^* has full rank. Form the Lagrangian

$$L(u, \lambda, v) = f(u) - \sum_{i=1}^m \lambda_i (g_i(u) - b_i) - \sum_{i=1}^n v_i [h_i(u_i) - c_i]$$

Then there exist multipliers $\lambda_1^*, \dots, \lambda_k^*$ and $v_1^*, \dots, v_m^*$ such that

$$\frac{\partial L}{\partial u_i}(u^*, \lambda^*, v^*) = 0 \text{ for all } i \in \{1, \dots, n\}$$

$$\lambda_i^* [g_i(u^*) - b_i] = 0 \text{ for all } i \in \{1, \dots, k\}$$

$$h_i(u^*) = c_i \text{ for all } i \in \{1, \dots, m\}$$

$$g_i(u^*) \leq b_i \text{ for all } i \in \{1, \dots, k\}$$

$$\lambda_i \geq 0 \text{ for all } i \in \{1, \dots, k\}$$

18.5.2 Kuhn-Tucker Formulations

The most common problems in economics are maximization problems dealing with only inequality constraints. Many of these constraints come in the form of non-negativity constraints, such as requiring consumption to be weakly positive. Consider the following problem:

$$\max_{u_i \in R} f(u_1, \dots, u_n)$$

Subject to

$$g_1(u_1, u_2, \dots, u_n) \leq b_1, \dots, g_m(u_1, \dots, u_n) \leq b_m$$

$$\text{And } x_1 \geq 0, \dots, x_n \geq 0$$

The Lagrangian function we would write would take the form

$$L(u, \lambda, v) = f(u) - \sum_{i=1}^m \lambda_i (g_i(u) - b_i) + \sum_{i=1}^n v_i u_i,$$

which would lead to the following first order conditions:

$$\frac{\partial L}{\partial u_1} = \frac{\partial f}{\partial u_1} - \sum_{i=1}^m \lambda_i \frac{\partial g_i}{\partial u_1} + v_1 = 0, \dots, \frac{\partial L}{\partial u_n} = \frac{\partial f}{\partial u_n} - \sum_{i=1}^m \lambda_i \frac{\partial g_i}{\partial u_n} + v_n = 0$$

$$\lambda_1[g_1(u) - b_1] = 0, \dots, \lambda_m[g_m(u) - b_m] = 0$$

$$g_1(u_1, u_2, \dots, u_n) \leq b_1, \dots, g_m(u_1, \dots, u_n) \leq b_m$$

$$v_1 u_1 = 0, \dots, v_n u_n = 0, \text{ and } \lambda_i u_j \geq 0, \forall i = 1, \dots, m \text{ and } \forall j = 1, \dots, n$$

These are a lot of conditions $(3m + 3n)$

Now consider the Lagrangian without the non-negativity constraints, and call it the Kuhn-Tucker Lagrangian:

$$L(u, \lambda, v) = \bar{L}(u, \lambda) + \sum_{i=1}^n v_i u_i$$

The first n first order conditions can be rewritten as

$$\frac{\partial L}{\partial u_j} = \frac{\partial \bar{L}}{\partial u_j} + v_j = 0, \quad \forall j = 1, \dots, n,$$

Which implies

$$\frac{\partial \bar{L}}{\partial u_j} = -v_j, \quad \forall j = 1, \dots, n.$$

Plugging those into the non-negativity constraints we have that

$$u_j \frac{\partial \bar{L}}{\partial u_j} = 0 \text{ and } \frac{\partial \bar{L}}{\partial u_j} \leq 0.$$

And notice that

$$\frac{\partial L}{\partial \lambda_j} = \frac{\partial \bar{L}}{\partial \lambda_j} = b_j - g_j(u) \geq 0, \quad \forall j = 1, \dots, m,$$

Which implies

$$\frac{\partial \bar{L}}{\partial \lambda_j} \geq 0 \text{ and } \lambda_j \frac{\partial \bar{L}}{\partial \lambda_j} = 0, \quad \forall j = 1, \dots, m.$$

In summary, the following are the first order conditions for the Kuhn-Tucker Lagrangian:

$$\frac{\partial \bar{L}}{\partial x_j} \leq 0 \text{ and } x_j \frac{\partial \bar{L}}{\partial x_j} = 0, \quad \forall j = 1, \dots, n$$

$$\frac{\partial \bar{L}}{\partial \lambda_j} \geq 0 \text{ and } \lambda_j \frac{\partial \bar{L}}{\partial \lambda_j} = 0, \quad \forall j = 1, \dots, m$$

$$\lambda_j \geq 0, \quad \forall j = 1, \dots, m$$

This is $2n + 3m$ constraints, n less than before. If we are dealing with a lot of non-negativity constraints it is therefore faster to use the Kuhn-Tucker Lagrangian.

Check Your Progress 3

1) Maximize $f(u_1, u_2) = u_1 u_2$ subject to $h(u_1, u_2) = u_1 + 4u_2 = 16$.

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- 2) Find the maximum of $f(u_1, u_2) = -(u_1 - 5)^2 - (u_2 - 5)^2$ subject to $u_1^2 + u_2 \leq 9$,
 $u_1, u_2 \geq 0$.

18.6 LET US SUM UP

The crucial concepts of convexity and concavity were thoroughly covered in this unit. Uses of concave and convex functions are common in economics, and they serve as a basis for the concept of optimisation, a fundamental and often occurring concept in economics. Further, the usage of second derivatives in relation to these functions was provided. Certain requirements and conditions of convex and concave functions along with the idea of quasi-concavity and quasi-convexity was also highlighted. The other part of the unit focused on the fundamentals of restricted optimisation, which can be broken down into two primary categories: constrained optimisation under equality constraints and constrained optimisation under inequality constraints. The complimentary slackness requirements that demonstrate that the primal and dual values are the same have also been presented in this unit. In addition, Kuhn-Tucker requirements for the general mathematical programming problem, such as non-negative and free variables, equality and inequality constraints, are included.

18.7 KEY WORDS

- Convex Function** : In mathematics, a real-valued function is called convex if the line segment between any two distinct points on the graph of the function lies above the graph between the two points.
- Differentiable Function** : A differentiable function of one real variable is a function whose derivative exists at each point in its domain. In other words, the graph of a differentiable function has a non-vertical tangent line at each interior point in its domain.
- Maximum and Minimum:** : A function, known generically as extremum (extrema), are the largest and smallest value taken by the function, either within a given

range (the *local* or *relative* extrema), or on the entire domain (the *global* or *absolute* extrema).

18.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISE

Check Your Progress 1

$$1) \text{ a) } H(u) = \begin{bmatrix} \frac{\partial^2 f}{\partial u_1^2} & \frac{\partial^2 f}{\partial u_1 \partial u_2} \\ \frac{\partial^2 f}{\partial u_1 \partial u_2} & \frac{\partial^2 f}{\partial u_2^2} \end{bmatrix} = \begin{bmatrix} 12u_1 & 0 \\ 0 & -12 \end{bmatrix}$$

Here $\frac{\partial^2 f}{\partial u_1^2} = 12u_1 \leq 0$ for $u_1 \leq 0$ and ≥ 0 and $u_1 \geq 0$, and $|H(u)| = -144u_1 \geq 0$

for $u_1 \leq 0$ and ≥ 0 and $u_1 \geq 0$. Hence $H(u)$ will be negative semi definite and $f(u)$ is concave for $u_1 \leq 0$.

b) $f(u) = -8u^2$, $H(u) = \frac{d^2 f}{du^2} = -16 < 0$ for all real values of u . Hence $f(u)$ is strictly concave.

2) The idea of quasi concavity is, in fact, less strong than the idea of concavity. This is due to the fact that while every quasi-concave function is also a concave function, the reverse is not always true.

Check Your Progress 2

1) The Lagrangian is

$$L = u_1^2 + u_2^2 + u_3^2 - \lambda_1(3u_1 + u_2 + u_3 - 5) - \lambda_2(u_1 + u_2 + u_3 - 1)$$

The first order conditions are

$$L_{u_1} = 2u_1 - 3\lambda_1 - \lambda_2 = 0$$

$$L_{u_2} = 2u_2 - \lambda_1 - \lambda_2 = 0$$

$$L_{u_3} = 2u_3 - \lambda_1 - \lambda_2 = 0$$

$$\text{And } L_{\lambda_1} = 3u_1 + u_2 + u_3 - 5 = 0$$

$$L_{\lambda_2} = u_1 + u_2 + u_3 - 1 = 0.$$

The linear system of five equations in five unknowns has a unique solution.

$\left[\left(2, \frac{-1}{2}, \frac{-1}{2} \right) \right]$. The Jacobian of the constraints is $\left[\begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \right]$, which has rank [2], and so the NDCQ holds.

2) The Lagrangian is

$$L = 3 + u_2 u_3 - \lambda(u_2^3 + u_3^2 - 1)$$

The first order conditions are

$$L_{u_2} = u_3 - 2\lambda u_2 = 0$$

$$L_{u_3} = u_2 - 2\lambda u_3 = 0$$

$$L_\lambda = u_2^2 + u_3^2 - 1 = 0$$

The solutions are the four (u_2, u_3) pairs such that $u_2 = \pm \frac{1}{\sqrt{2}}$ and $u_3 = \pm \frac{1}{\sqrt{2}}$. For a maximum, u_2 and u_3 must have the same sign, so the solutions are $(3\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $(-3\sqrt{2}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$.

The value of the maximum in each case is $\frac{7}{2}$, and NDCQ holds.

3) Solve as per 1) and 2)

Check Your Progress 3

1) Form the Lagrangian $L(u_1, u_2) = u_1 u_2 - \lambda(u_1 + 4u_2 - 16)$

The first order conditions are

$$\frac{dL}{du_1} = u_2 - \lambda = 0$$

$$\frac{dL}{du_2} = u_1 - 4\lambda = 0$$

$$\frac{dL}{d\lambda} = u_1 + 4u_2 - 16 = 0$$

From the first two equations we have

$$u_1 = 4u_2$$

Plugging this into the last equation we have that $u_2 = 2$, which in turn implies that $u_1 = 8$, and that $\lambda = 2$. This states that the only candidate for the solution is $(u_1, u_2, \lambda) = (8, 2, 2)$

First let us form the Jacobian matrix

$$Dh(u_1, u_2, u_3) = \begin{pmatrix} 2u_1 & 2u_2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Notice that this is only singular if both $u_1 = u_2 = 0$. However, if this is the case, then our first constraint fails to hold. Therefore, the matrix is non-singular for all points in the domain, and so we do not need to worry about the NDCQ. Form the Lagrangian

$$L(u_1, u_2, u_3) = u_1 u_2 u_3 - \lambda_1 (u_1^2 + u_2^2 - 1) - \lambda_2 (u_1 + u_3 - 1)$$

The first order conditions are

$$\frac{dL}{du_1} = u_2 u_3 - 2\lambda_1 u_1 - \lambda_2 = 0$$

$$\frac{dL}{du_2} = u_1 u_3 - 2\lambda_1 u_1 = 0$$

$$\frac{dL}{du_3} = u_1 u_2 - \lambda_2 = 0$$

$$\frac{dL}{d\lambda_1} = u_1^2 + u_2^2 - 1 = 0$$

$$\frac{dL}{d\lambda_2} = u_1 - u_3 - 1 = 0$$

We can solve the second and third equations for λ_1 and λ_2

$$\lambda_1 = \frac{u_1 u_3}{2u_2}, \quad \lambda_2 = u_1 u_2$$

and plug these into the first equation to get

$$u_2 u_3 - 2 \frac{u_1^2 u_3}{2u_2} - u_1 u_2 = 0$$

Multiply both sides by u_2

$$u_2^2 u_3 - u_1^2 u_3 - u_1 u_2^2 = 0$$

Now we want to solve the two constraints for u_2 and u_3 in terms of u_1 , plug them into this equation, and get one equation in terms of u_1

$$(1 - u_1^2)(1 - u_1) - u_1^2(1 - u_1) - u_1(1 - u_1^2) = 0$$

$$(1 - u_1)[(1 - u_1^2) - u_1^2 - u_1(1 + u_1)] = 0$$

$$(1 - u_1)[1 - 3u_1^2 - u_1] = 0$$

This yields, $u_1 = \left\{ 1, \frac{-1 + \sqrt{13}}{6}, \frac{-1 - \sqrt{13}}{6} \right\}$. Let us first analyse

$u_1 = 1$. From the second constraint we have that $u_3 = 0$, and from the first constraint we have that $u_2 = \{1, -1\}$. However, this violates the equation

$$u_2^2 u_3 - u_1^2 u_3 - u_1 u_2^2 = 0, \text{ and therefore } u_1 = 1 \text{ cannot be a solution.}$$

Plugging in the other values, we get four candidate solutions

$$u_1 = .4343, \quad u_2 = \pm 0.9008, \quad u_3 = 0.5657$$

$$u_1 = -.7676, \quad u_2 = \pm 0.6409, \quad u_3 = 1.7676$$

2) Lagrange function:

$$L(u_1, u_2; \lambda) = -(u_1 - 5)^2 - (u_2 - 5)^2 + \lambda(9 - u_1^2 - u_2)$$

Kuhn- Tucker Conditions

$$(A) \quad L_{u_1} = -2(u_1 - 5) - 2\lambda u_1 \leq 0$$

$$(B) \quad L_{u_2} = -2(u_2 - 5) - \lambda \leq 0$$

$$(C) \quad L_\lambda = 9 - u_1^2 - u_2 \geq 0$$

$$(D) \quad u_1, u_2, \lambda \geq 0$$

$$(1) \quad u_1 L_{u_1} = -u_1(2(u_1 - 5) + 2\lambda u_1) = 0$$

$$(2) \quad u_2 L_{u_2} = -u_2(2(u_2 - 5) + \lambda) = 0$$

$$(3) \quad \lambda L_\lambda = \lambda(9 - u_1^2 - u_2) = 0$$

Express equations (1) - (3) as

$$(1) \quad u_1 \geq 0 \text{ or } 2(u_1 - 5) + 2\lambda u_1 = 0$$

$$(2) \quad u_2 \geq 0 \text{ or } 2(u_2 - 5) + \lambda = 0$$

$$(3) \quad \lambda = 0 \text{ or } 9 - u_1^2 - u_2 = 0$$

We have to compute all 8 combinations and check whether the resulting solutions satisfy inequalities (A), (B), (C) and (D).

If $\lambda = 0$ then from (1) and (2) there exist four solutions for $(u_1, u_2; \lambda)$:

$$(0, 0; 0), (0, 5; 0), (5, 0; 0) \text{ and } (5, 5; 0).$$

However, none of these points satisfies all inequalities (A), (B), (C).

Hence $\lambda \neq 0$.

If $\lambda \neq 0$ then from (3) implies $u_2 = 9 - u_1^2$.

If $\lambda \neq 0$ and $u_1 = 0$, then $u_2 = 9$ and because of (2), $\lambda = -8$. A contradiction to (D).

If $\lambda \neq 0$ and $u_2 = 0$, then $u_1 = 3$ and because of (1), $\lambda = \frac{2}{3}$. A contradiction to (B).

Consequently, all three variables must be non-zero. Thus $u_2 = 9 - u_1^2$ and

$\lambda = -2(u_2 - 5) = -2(4 - u_1^2)$. Substituted in (1) gives $2(u_1 - 5) - 4(4 - u_1^2)u_1 = 0$ and

$$u_1 = \frac{\sqrt{11} + 1}{2} \approx 2.158, \quad u_2 = \frac{12 - \sqrt{11}}{2} \approx 4.342, \quad \lambda = \sqrt{11} - 2 \approx 1.317$$

The Kuhn-Tucker conditions are thus satisfied only in point

$$(u_1, u_2; \lambda) = \left(\frac{\sqrt{11} + 1}{2}, \frac{12 - \sqrt{11}}{2}; \sqrt{11} - 2 \right).$$

18.9 EXERCISES

Q1. Maximise $f(u_1, u_2) = u_2$

Subject to

$$g_1(u_1, u_2) = u_1^2 + u_2^2 - 4 \leq 0$$

$$g_2(u_1, u_2) = -u_1^2 + u_2 \leq 0$$

$$g_3(u_1, u_2) = -u_1 \leq 0$$

$$g_4(u_1, u_2) = -u_2 \leq 0$$

Ans. The Kuhn Tucker conditions are given by

$$u_1^2 + u_2^2 - 4 \leq 0, -u_1 \leq 0$$

$$-u_1^2 + u_2 \leq 0, -u_2 \leq 0$$

$$2u_1\lambda_1 - 2u_1\lambda_2 - \lambda_3 = 0$$

$$2u_2\lambda_1 + \lambda_2 - \lambda_4 = 1$$

$$\lambda_1(u_1^2 + u_2^2 - 4) = 0$$

$$\lambda_2(-u_1^2 + u_2) = 0$$

$$\lambda_3 u_1 = 0, \lambda_4 u_2 = 0, \lambda_1 \geq 0, \lambda_2 \geq 0, \lambda_3 \geq 0, \lambda_4 \geq 0$$

The optimal solution can be obtained by inspection as

$$u_1 = 1.24962, u_2 = 1.56155, \lambda_1 = \lambda_2 = 0.24254, \lambda_3 = \lambda_4 = 0.$$

This solution satisfies the Kuhn Tucker condition. But the solution $u_1 = 0, u_2 = 0, \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = \lambda_4 = 0$ which is not optimum also satisfies the Kuhn Tucker conditions.

Q2. Show whether it is convex, concave or neither;

I) $f(u) = 10 - u^2$

Ans. $f(u) = 10 - u^2$

$$\frac{df}{du} = 0 - 2u \quad \text{and} \quad \frac{d^2f}{du^2} = -2$$

Since $\frac{d^2f}{du^2}$ is always < 0 for all value of u , the function is concave.

II) $f(u_1, u_2) = 5u_1 + 2u_2^2$

Ans. $\frac{df}{du_1} = 5$ and $\frac{\partial^2 f}{\partial u_1^2} = 0,$

$$\frac{df}{du_2} = 4u_2, \quad \frac{\partial^2 f}{\partial u_2^2} = 4 > 0,$$

$$\frac{\partial^2 f}{\partial u_1 \partial u_2} = 0.$$

$$\frac{\partial^2 f}{\partial u_1^2} = 0, \quad \frac{\partial^2 f}{\partial u_1^2} = 0 - \left[\frac{\partial^2 f}{\partial u_1 \partial u_2} \right]^2 = 0.$$

The function is convex.

Q3. Verify whether the following function is convex or concave?

$$f(u) = 4u_1^3 + 3u_2^2 + u_3^2 - 6u_1u_2 + u_1u_3 - \frac{u_1}{2} - 2u_2 + 15.$$

Ans. The necessary conditions are

$$\frac{df}{du_1} = 8u_1 - 6u_2 + u_3 - \frac{1}{2} = 0,$$

$$\frac{df}{du_2} = 6u_2 - 6u_1 - 2 = 0,$$

$$\frac{df}{du_3} = 2u_3 + u_1 = 0.$$

The solution of these equations gives $u_1 = -\frac{5}{3}, u_2 = -2, u_3 = -\frac{5}{6}$.

$$H(u) = \begin{bmatrix} \frac{\partial^2 f}{\partial u_1^2} & \frac{\partial^2 f}{\partial u_1 \partial u_2} & \frac{\partial^2 f}{\partial u_1 \partial u_3} \\ \frac{\partial^2 f}{\partial u_2 \partial u_1} & \frac{\partial^2 f}{\partial u_2^2} & \frac{\partial^2 f}{\partial u_2 \partial u_3} \\ \frac{\partial^2 f}{\partial u_3 \partial u_1} & \frac{\partial^2 f}{\partial u_3 \partial u_2} & \frac{\partial^2 f}{\partial u_3^2} \end{bmatrix} = \begin{bmatrix} 8 & -6 & 1 \\ -6 & 6 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

The principal minor determinants are $|8|, \begin{vmatrix} 8 & -6 \\ -6 & 6 \end{vmatrix} = 12, \begin{vmatrix} 8 & -6 & 1 \\ -6 & 6 & 0 \\ 1 & 0 & 2 \end{vmatrix} = 18$.

Since all the principal minor determinants are non - negative, the function is convex.